Name	$M\sharp$

Models

Review $y' = A\,y + b$

1. Write down the complementary solution $y_c = \sum_{i=1}^{n} c_i\, y_i$ satisfying $y_c{}' = A\, y_c$

1.1. One solution when λ is real and $A\,v = \lambda\,v$

$e^{\lambda t}\, v$

1.2. Two solutions when $\lambda = a \pm b\,i$ and $A\,(p + q\,i) = (a + b\,i)\,(p + q\,i)$

$e^{a t}(\cos(b\,t)\,p - \sin(b\,t)\,q)$ and $e^{a t}(\sin(b\,t)\,p + \cos(b\,t)\,q)$

1.3. Two solutions when λ is repeated twice but there is only one eigenvector $A\,v = \lambda\,v$

$e^{\lambda t}\,v$ and $t\,e^{\lambda t}\,v + e^{\lambda t}\,u$

where u satisfies $(A - \lambda\, I)\,u = v$

2. Compute a particular solution y_p satisfying $y_p{}' = A\,y_p + b$

2.1. Guess the form (polynomial in t, exponentials $e^{\omega t}$, $\sin(\omega\,t)$, …) for y_p if you can and compute the coefficients by substituting into the ODE.

2.2. Use variation of parameters $y = \Phi(t)\,u(t)$ where

$u(t) = \int^{t} \Phi(\tau)^{-1}\, b(\tau)\, d\tau + c$

if you can not guess the form.

3. Write down the general solution $y = y_c + y_p$.

4. Substitute initial conditions (if there are) to determine the vector $c = [c_1, c_2, \ldots, c_n]$.

Example

Solving higher order constant coefficient ODEs e.g.

$$x'' + 2\,x' + x = t$$

We can convert to a system by introducing $v = x'$ like this

$$x'' = -2\,x' - x + t$$

so rearranging gives

$$\begin{aligned} x' &= v \\ v' &= -x - 2\,v + t \end{aligned}$$

which with $y = \begin{pmatrix} x \\ v \end{pmatrix}$ is

$$y' = \begin{pmatrix} 0 & 1 \\ -1 & -2 \end{pmatrix} y + \begin{pmatrix} 0 \\ t \end{pmatrix}$$

The matrix has one eigen vector $v = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$ for a repeated eigenvector of $\lambda = -1$ so the complementary solution is

$$y_c = c_1 e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 e^{-t} \left(t \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right)$$

In[6]:=
```
MatrixForm[Eigensystem[A = ( 0   1
                            -1  -2 )]]
LinearSolve[A - (-1) IdentityMatrix[2], {-1, 1}]
```

Out[6]//MatrixForm=
```
(    -1         -1
  {-1, 1}    {0, 0} )
```

Out[7]=
```
{-1, 0}
```

A particular solution is of the form

$$y_p = u_0 + t\, u_1$$

where the undetermined coefficients satisfy

$$u_1 = A(u_0 + t\, u_1) + t \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Matching coefficients gives

$$0 \;=\; A\, u_1 \;+\; \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$u_1 \;=\; A\, u_0 \;+\; \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

rewriting gives

$$\begin{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} & -A \\ -A & \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{pmatrix} \begin{pmatrix} u_0 \\ u_1 \end{pmatrix} = \begin{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \end{pmatrix}$$

In[8]:=
```
LinearSolve[ArrayFlatten[(  0            -A
                           -A  IdentityMatrix[2] )], {0, 1, 0, 0}]
```

Out[8]=
```
{-2, 1, 1, 0}
```

So $u_0 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$ and $u_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and the general solution is

$$y = \begin{pmatrix} x \\ v \end{pmatrix} = y_c + y_p = c_1 e^{-t} \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 e^{-t} \left(t \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \end{pmatrix} \right) + \begin{pmatrix} -2 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

We might only care about

$$x(t) = -c_1 e^{-t} - c_2 (1 + t) e^{-t} - 2 + t$$

The last set of topics are how to solve higher order constant coefficient linear ODEs directly without converting them to a system! This is Chapter 4.

Exam 2 Tuesday

The coverage for the exam is:

1. Question 1 is a Linear ODE from the 1st part of the course.
2. Writing down real solutions of $y' = A\,y$ when given eigenvalues and eigenvectors. This includes.
   2.1. Real eigenvalues
   2.2. Complex conjugate pairs of eigenvalues $a \pm b\,i$
   2.3. Missing eigenvectors
3. Writing down particular solution forms for $y' = A\,y + b$ and writing down the linear systems which determine the coefficients. The forms (with include a_j vectors of constants)
   3.1. $b = e^{\omega t}\,a_0$
   3.2. $b = \sin(\omega\,t)\,a_0$ and $b = \cos(\omega\,t)\,a_0$
   3.3. $b = a_0 + a_1\,t + a_2\,t^2 + \ldots$
4. Writing down the linear system that determines the constants of integration from an initial condition.

The exam has five ten point questions on it. It will be on two sides of two sheets of paper. You are allowed a calculator and a note card. The exam will be scanned then graded and returned using Gradescope.